Certifying Reconstruction from Finite Data

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Abstract

Papers [6, 5] establish that a σ-additive probability measure is forced by observational coherence (collective exhaustion), and that this measure already contains, without further assumption, both the conditional regularity structure and the dynamical semigroup — with reconstruction as the question of whether the minimal sufficient factor is faithful. This paper asks what finite data can certify about those structures.

At the level of a general query system, the *population separation defect* $\delta(\mathcal{G})$ measures how far finite observations fall from full resolution; it vanishes under collective exhaustion and honest refinement. The empirical collision probability is a U-statistic proxy, and a persistent empirical floor is evidence against collective exhaustion — a CE-shadow theorem requiring no temporal or geometric structure.

In the dynamical specialisation, the certification framework meets the reconstruction theorem of [5]: honest refinement, a design condition in the general setting, is here automatic — a consequence of reconstruction — and the remaining degrees of freedom (mixing rate, embedding dimension) are what rate-optimal guarantees exploit. Three theorems certify reconstruction from a time series $x_1, \ldots, x_n$. The *Algebra Theorem* shows that the elbow stopping rule achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without oracle knowledge of the mixing rate, lag, or smoothness index. The *Dynamics Theorem* shows that under uniform separation, empirical delay vectors constitute a geometric witness converging at rate $n^{-\beta/(2\beta+d)}$. The *Conjunction Theorem* shows that for deterministic T, the algebraic and geometric witnesses certify the same event — not merely correlated, but identical.

All three witnesses, together with a fourth information-theoretic one from the collision entropy of the delay-vector distribution, measure the same failure: separation collapsing across delay fibres. Under fibre mixing, they are equivalent. The convergence of three independently motivated certifications onto a single geometric object is the main message.

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1 Introduction

A probability measure on an observable algebra already contains, without further assumption, a conditional regularity structure — the systematic, outcome-by-outcome account of how one observation is distributed given another [5]. Concretely, the Rokhlin disintegration of any two observations Q, F gives a Markov kernel $\kappa_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$ that is forced by the measure, not chosen. Composing with Q gives the minimal sufficient factor Q_*, the coarsest reduction of the sample point carrying full conditional information about F. In a measure-preserving system with $Q_t = h \circ T^t$, this factor becomes the delay map Φ_h, and the reconstruction question becomes internal: is the factor faithful, or does it see only a proper quotient of X? The reconstruction theorem of [5] answers in terms of an observable algebra condition — $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ generates $\mathcal{B}$ modulo μ if and only if Φ_h is a measure-theoretic embedding. This is an exact asymptotic result; it says nothing about rates, certificates, or what can be inferred from a finite time series. The results of [6, 5] are taken as given; this paper asks what finite data can certify about the structure they establish.

The certification question has two natural levels, and the paper works from the more general inward. At the level of a general query system [6] — before any dynamical or geometric structure is assumed — one can ask: given finite responses to a directed family of queries, does the empirical evidence certify that the observable structure is resolving correctly, and can CE-failure be detected from data? Section 2 answers this at full generality. The central objects are the population separation defect $\delta(\mathcal{G})$ and its empirical U-statistic proxy; the main results are a population convergence theorem (CE drives the defect to zero under honest refinement), an estimation theorem (the empirical proxy concentrates under IID sampling), and a CE-shadow corollary (a persistent empirical floor is evidence against CE). The corollary has no temporal hypothesis; the admissibility condition CE acquires an empirical signature that is entirely independent of dynamics.

Moving to the dynamical specialisation, the two threads of the programme converge. Section 2's certification framework meets the reconstruction theorem of [5], and the meeting point is stronger than either alone. In the general query setting, honest refinement is a design condition: it must be built into the query family. In the dynamical setting, it is automatic — a consequence of the reconstruction theorem itself, which guarantees that $\sigma(\bigcup_L \mathcal{O}_h^{(L)}) = \mathcal{B}$ modulo μ once the factor is faithful. The observer does not need to design an honest query path; the dynamics provide it. The remaining degrees of freedom — mixing rate and embedding dimension — are what the rate theorems exploit. The defect specialises to $\delta(L) = \delta(\mathcal{O}_h^{(L)})$. Three theorems certify reconstruction in this regime.

The Algebra Theorem (Section 6). The elbow stopping rule $\hat{L}^*$ fires when the empirical defect $\hat{\delta}(L, n)$ stops decreasing. Under exponential mixing with resolvability constant $C_\lambda > 4$, $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d-dimensional state space, without any oracle knowledge of the mixing rate, the lag $L^*(n)$, or the smoothness index s.

The Dynamics Theorem (Section 7). When $T \in C^r$ and the uniform separation condition (US) holds, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz, and the empirical delay vectors constitute a second witness certifying that points are being separated by the dynamics. This kernel witness converges at rate $n^{-\beta/(2\beta+d)}$ and fails precisely when (US) fails.

The Conjunction Theorem (Section 8). For deterministic T, the TV separation $d_L(x, x') = 1$ if and only if x and x' are separated by $\mathcal{O}_h^{(L)}$ — the algebraic identity $\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$ is the Markov bridge. Under reconstruction and (US), both witnesses certify the same object; the theorem names two failure modes (reconstruction failure collapses the algebra witness; (US)-failure collapses the kernel witness) that jointly exhaust the ways certification can fail.

The discussion section records where each proof requires mixing and what would be needed to extend beyond it.

2 General observational certification

Fix a query system $(\mathcal{E}_n)_n$ with observable σ-algebra $\sigma(\mathcal{C})$ and realization measure μ satisfying CE [6]. For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \cdots \vee \mathcal{E}_{n_k}$, the *population separation defect* is

$$\delta(\mathcal{G}) \ := \ \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E).$$

It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C})$ mod μ. A scheme $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C})$ mod μ; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Theorem 2.1 (Population theorem)**.** *Under CE and any honest refinement scheme* $(\mathcal{G}_k)$*,*

$$\delta(\mathcal{G}_k) \ \longrightarrow \ 0.$$

Proof. Since $(\mathcal{G}_k)$ is honest, every $A \in \sigma(\mathcal{C})$ is $\sigma(\bigcup_k \mathcal{G}_k)$-measurable, so the martingale $\mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ converges to $\mathbf{1}_A$ in $L^1(\mu)$. Thus $\inf_{E\in\mathcal{G}_k} \mu(A \triangle E) \to 0$ for each A. CE [6] ensures this convergence is uniform: no mass persists on the residual sets. Hence $\delta(\mathcal{G}_k) \to 0$. $\square$

Given N independent query responses $\omega_1, \dots, \omega_N$ drawn from μ, let $R_i = (Q_{n_1}(\omega_i), \dots, Q_{n_k}(\omega_i)) \in \alpha_{n_1} \times \cdots \times \alpha_{n_k}$ be the joint response vector of the i-th sample. Define the *empirical separation defect*

$$\hat{\delta}(\mathcal{G}, N) \;:=\; \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \mathbf{1}[R_i = R_j],$$

the empirical collision probability of the joint response. This is a second-order U-statistic requiring no geometric structure on Ω.

Theorem 2.2 (Estimation theorem)**.** *Under IID sampling from the observable law of the joined query responses, $\hat{\delta}(\mathcal{G}, N) \to \delta(\mathcal{G})$ in probability as $N \to \infty$.*

Proof. $\hat{\delta}(\mathcal{G}, N)$ is a second-order U-statistic for a bounded kernel; the result is an instance of the standard U-statistic law of large numbers [3]. $\square$

Corollary 2.3 (CE-shadow)**.** *Under an honest refinement scheme and IID sampling, if $\hat{\delta}(\mathcal{G}_k, N) \geq c > 0$ for all large k and N, that is evidence against CE: CE is what guarantees the floor can be discharged by further refinement.*

Remark 2.4 (Scope)**.** The corollary is one-directional: persistent floor implies CE-failure; $\delta(\mathcal{G}_k) \to 0$ does not imply CE. The defect is defined inside $\sigma(\mathcal{C})$, while CE also controls escape to the Stone boundary — non-principal ultrafilters invisible to the defect functional. A charge family can resolve $\sigma(\mathcal{C})$ completely while still placing mass on non-realised boundary points [6].

Honest refinement is essential: a stalling empirical defect under a poorly-chosen query path is a design artifact, not an admissibility failure. Under an honest scheme it is the latter.

CE thereby acquires an empirical signature absent from its abstract treatment in [6]: it is not only the admissibility condition for σ-additivity, but a structural hypothesis with a detectable finite-sample consequence.

The remainder of this paper establishes what becomes achievable when the general framework of this section meets the dynamical structure of [5]. Specialisation to a dynamical setting might seem to narrow the question; instead it strengthens it. Honest refinement, which must here be a design condition, becomes free; the separation defect, which is abstract, becomes geometrically interpretable; and the witnesses, which here are qualitative (floor or no floor), become quantitative with optimal rates. That gain is the subject of Sections 6–8.

3 Setup and Notation

The measure-preserving system. Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d-dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \to \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} \;:=\; \operatorname{essinf}_X \rho \;>\; 0.$$

Let $T : X \to X$ be a μ-preserving measurable map. We make no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

The observable and the delay map. Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \to \mathbb{R}^{L+1}, \qquad \Phi_h^{(L)}(x) \;=\; \big(h(x),\, h(Tx),\, \dots,\, h(T^L x)\big).$$

The *observable σ-algebra at lag L* is

$$\mathcal{O}_h^{(L)} \ := \ \sigma(\Phi_h^{(L)}) \ = \ \sigma\{h \circ T^k : 0 \le k \le L\}.$$

The *delay-polynomial class* of degree $D \ge 1$ is

$$\mathcal{A}_h^{(L)} \ := \ \{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree} \le D \text{ on } \mathbb{R}^{L+1}\} \ \subseteq \ L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

The approximation error. The central quantity is the σ-algebra approximation error

$$\delta(L) \ := \ \delta(\mathcal{O}_h^{(L)}) \ = \ \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E), \tag{1}$$

which measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. It is non-increasing in L, vanishes if and only if reconstruction holds, and under (G1)–(G2) on a smooth manifold reaches zero by lag $L_0 \le d - 1$ [5].

Fibre structure and collision probability. The delay map $\Phi_h^{(L)}$ partitions X into *fibres*: for each $z \in \mathbb{R}^{L+1}$, define

$$F_z \ := \ (\Phi_h^{(L)})^{-1}(z), \qquad p_z \ := \ \mu(F_z), \qquad \mu_z \ := \ \mu(\,\cdot \mid F_z).$$

Call z an *ε-large fibre* if $p_z \ge \varepsilon$. The push-forward $\nu_L := (\Phi_h^{(L)})_*\mu$ is the distribution of delay vectors.

The set of unseparated pairs $R_L := \{(x, x') : \Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')\}$ decomposes as $R_L = \bigsqcup_z F_z \times F_z$, giving

$$(\mu \otimes \mu)(R_L) \ = \ \int p_z^2 \, d\nu_L(z). \tag{2}$$

This quantity is the *collision probability* of the delay-vector distribution: it is the probability that two independent draws from μ produce the same delay vector.

Remark 3.1 (Information horizon)**.** The fibre structure imposes a hard constraint on what finite data can certify. At lag L the observable algebra partitions X into at most 2^{L+1} atoms (in the binary-coded case; more generally, the number of fibres grows exponentially with L). With a sample of size n, the mean occupancy per fibre is $n/2^{L+1}$, which halves with each additional lag. The *information horizon* is

$$L^* \ := \ \lfloor \log_2 n \rfloor,$$

the lag at which mean fibre occupancy drops to 1. Beyond L^* the sample is partitioned across more fibres than it contains points, so empirical fibre statistics become unreliable. This gives a first-principles derivation — from the observable algebra structure alone — of why the concentration rate $\varepsilon_n \sim n^{-2s/(2s+d)}$ and the elbow stopping rule of Theorem 8.3 fire when they do: doubling the sample size buys exactly one more trustworthy lag.

The data model. We observe a time series $x_1, x_2, \ldots, x_n$ which is an orbit segment of T: $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ. The empirical measure is $\mu_n := \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$. We do not assume ergodicity or density of the orbit; each result states the mixing hypothesis it uses.

Structural hypotheses. The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

(G1) *No short periodic orbits:* T has no periodic orbit of period $\le L$ in $\mathrm{supp}(\mu)$. This ensures the orbit points $x, Tx, \ldots, T^L x$ are distinct μ-a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.

(G2) *Non-degenerate observable:* $h \in C^2(X)$ with $dh \neq 0$ μ-a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.

(US) *Uniform separation:* $\inf_{x\in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ-a.e. Condition (US) is used only in the Dynamics and Conjunction theorems (Sections 7 and 8).

The empirical witness. The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) \;:=\; \sup_{p\in\mathcal{P}_D^{(L)},\, \|p\|_{L^2(\mu)}\le 1} \big\|p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]\big\|^2_{L^2(\mu_n)},$$

where $\mathcal{P}_D^{(L)}$ is the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 5.5).

The main result of Section 5 is that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C\, n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

4 Bias Bound and Conditional Variance

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub-σ-algebra, with $\delta(\mathfrak{m})$ as in (1).

Lemma 4.1 (Conditional variance identity)**.** *For any $S \in \mathcal{B}$:*

$$\mathbb{E}\big[\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})\big] \;=\; \frac{1}{2}\int_{R_L} \big|\mathbb{1}_S(x) - \mathbb{1}_S(x')\big|^2\, d(\mu\otimes\mu)(x, x').$$

Proof. By disintegration, $(\mu\otimes\mu)|_{R_L} = \int \mu_z \otimes \mu_z \, d\nu_L(z)$. On each fibre F_z:

$$\tfrac{1}{2}\int_{F_z\times F_z} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2\, d(\mu_z\otimes\mu_z) \;=\; \mu_z(S)\,\mu_z(S^c).$$

Integrating with weight p_z: $\frac{1}{2}\int_{R_L} |\mathbb{1}_S - \mathbb{1}_{S'}|^2\, d(\mu\otimes\mu) = \int p_z\, \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. Since $\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})(x) = \mu_z(S)\mu_z(S^c)$ on F_z, both sides equal $\int \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. ☐

Bounding $|\mathbb{1}_S - \mathbb{1}_{S'}|^2 \le 1$ in the lemma and taking the supremum gives $\delta(L) \le \frac{1}{2}(\mu\otimes\mu)(R_L)$.

Remark 4.2 (Finite analogue of collective exhaustion)**.** This bound and its converse (Corollary 6.12, proved under fibre mixing in §6) together say: algebraic indistinguishability $\delta(L)$ vanishes if and only if the mass of indistinguishable pairs $(\mu\otimes\mu)(R_L)$ vanishes. This is the finite-sample counterpart of collective exhaustion [6]. There, CE asserts that set-theoretic vanishing must be witnessed at some finite level of the refinement order. Here, the analogous principle is that observational indistinguishability cannot persist globally without manifesting as a non-vanishing mass of indistinguishable pairs at finite lag L. The fibre mixing condition plays an analogous structural role to CE: it is a sufficient condition under which the global and pairwise witnesses are comparable. Whether fibre mixing is itself derivable from dynamical conditions — or is genuinely irreducible, as CE is proved to be in [6] — remains open; see the open problems in Section 9.

Lemma 4.3 (Bias bound)**.** *For every $f \in L^\infty(\mu)$,*

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty}\cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. For an indicator $\mathbb{1}_S$: the conditional expectation is the L^2-projection, so $\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \leq \mu(S \triangle E) \leq \delta(\mathfrak{m})$ for the best $E \in \mathfrak{m}$. The general bounded case follows by the layer-cake representation $f = \int_{-M}^{M} \mathbb{1}_{\{f>t\}}\, dt$ and Minkowski's inequality for integrals, giving $\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} \leq 2M \cdot \delta(\mathfrak{m})^{1/2}$. $\square$

In the delay setting of Section 3, let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 4.4 (Three-term decomposition)**.** *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{bias} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{approx.} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{statistical}, \quad (3)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (1) *and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.*

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 4.3. $\square$

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n.

Lemma 4.5 (Observable reverse lower bound)**.** *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 *as $n \to \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in* (1) *at lag L and $\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ is the empirical measure.*

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \to 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2-projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

5 Concentration of the Empirical Witness

Throughout this section: (T, h) satisfies (G1)–(G2), X is a compact smooth d-manifold, μ has smooth positive density, and $d > 2s$.

5.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \le 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 5.1 (Uniform boundedness)**.** *Define*

$$M_{D,L} := \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \operatorname{essinf}_X \rho > 0$ *is the infimum of the density of* μ*. Then for every* $p \in \mathcal{P}_D^{(L)}$*:*

$$\|p\|_{L^\infty} \le M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to $y = x/a \in [-1,1]^{L+1}$ and expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \le D}$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives $|\tilde{p}(y)| \le N(D, L+1)^{1/2} (\sum |c_\alpha|^2)^{1/2}$ with $N(D, L+1) = \binom{D+L+1}{L+1}$. The Legendre orthogonality $\|P_\alpha\|_{L^2}^2 = \prod_i (2\alpha_i + 1)^{-1}$ gives $\sum |c_\alpha|^2 \le (2D+1)\|\tilde{p}\|_{L^2}^2$ (the maximum $(2D+1)$ is L-independent, achieved at $\alpha = (D, 0, \ldots, 0)$). Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\mathrm{vol})}$ and $\|p\|_{L^2(\mathrm{vol})} \le \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. □

5.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} = \Big\{ x \mapsto \big(p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x)\big)^2 \ : \ p \in \mathcal{P}_D^{(L)},\ \|p\|_{L^2(\mu)} \le 1 \Big\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 5.5). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 5.2 (Concentration, $\star$)**.** *Assume* $d > 2s$*. There exist constants* $c, C > 0$ *depending only on* $(d, s, \|h\|_{L^\infty})$ *such that for all* $t > 0$ *and all* $n \ge 1$*:*

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \le 2 \exp\Big(-\frac{c\, n\, t^2}{M_{D,L}^4 \cdot D^d}\Big). \qquad (\star)$$

In particular, setting $t = \varepsilon_n/2$ *with* $\varepsilon_n = C'\, n^{-2s/(2s+d)}$*:*

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > \varepsilon_n/2\big) \le 2 \exp\Big(-c'\, n^{(d-2s)/(2s+d)}\Big) \to 0.$$

Proof. Uniform boundedness of $\mathcal{F}_{L,D}$ (Lemma 5.1) gives $\|f\|_{L^\infty} \le 4M_{D,L}^2$ for each $f \in \mathcal{F}_{L,D}$, so a one-point replacement changes $\hat{\delta}(L,n)$ by at most $4M_{D,L}^2/n$. The McDiarmid inequality [7] gives the stated exponential tail. The bias term $|\mathbb{E}[\hat{\delta}(L,n)] - \delta(L)|$ is controlled by symmetrisation and the Ledoux–Talagrand contraction lemma [4], reducing to the Rademacher complexity of $\mathcal{P}_D^{(L)}$; since $\Phi_h^{(L)}(X)$ is a d-dimensional submanifold in the Takens regime, the VC dimension of $\mathcal{P}_D^{(L)}$ is $O(D^d)$ [2] and $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \le C\sqrt{D^d/n}$. Combining and setting $t = C' n^{-2s/(2s+d)}$ gives the result; the exponent diverges iff $d > 2s$. □

Remark 5.3 (The $d > 2s$ hypothesis)**.** When $d \le 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [1] is technically available but left to future work; we take $d > 2s$ as a standing hypothesis.

5.3 Source 1: the LLN gap

Lemma 5.4 (LLN gap). *With notation as above:*

$$\sup_{p\in\mathcal{P}_D^{(L)}}\left|\|p-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu_n)}-\|p-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu)}\right|\ \le\ C\sqrt{D^d/n}$$

almost surely as $n\to\infty$, where C depends on $(d,D,M_{D,L})$.

Proof. The class $\{(p-\mathbb{E}[p|\Phi_h^{(L)}])^2 : p\in\mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [2] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. □

5.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 5.5 (Nadaraya–Watson estimator). For bandwidth $h_n>0$ and kernel $K:\mathbb{R}^{L+1}\to\mathbb{R}_{\ge0}$ (compactly supported, symmetric, $\int K=1$):

$$\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}(x)]\ :=\ \frac{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)\,p(x_j)}{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)}.$$

Proposition 5.6 (Source 2 estimator bias). *Assume:*

(i) $T,h\in C^r$ with $r\ge2$, μ smooth positive density ρ with $\rho_{\min}>0$.

(ii) $\Phi_h^{(L)}$ has full rank μ-a.e. (follows from (G1)–(G2) and Lemma 6.2).

(iii) T is exponentially mixing with rate $\lambda>0$ and $\|h\|_{L^\infty}\ge\frac12$.

(iv) The NW bandwidth satisfies $h_n\sim n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ with $d_{\mathrm{eff}}(L)=\min(L+1,d)$ and $\beta\ge1$.

Then there exists $C_2(D,L)>0$ such that for all $p\in\mathcal{P}_D^{(L)}$:

$$\mathbb{E}\|\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}]-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu)}\ \le\ C_2(D,L)\cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))},$$

uniformly in $p\in\mathcal{P}_D^{(L)}$, where $C_2(D,L)=2M_{D,L}^2C_1^2A^{2\beta}$ and A is a constant depending on $(\beta,d_{\mathrm{eff}}(L),\rho_{\min})$.

Proof. The key input is that the conditional expectation $z\mapsto\mathbb{E}[p\mid\Phi_h^{(L)}=z]$ is Hölder(β) uniformly over $p\in\mathcal{P}_D^{(L)}$. This follows from TV-Hölder continuity of the fibre measures: under conditions (i)–(ii), a Rokhlin disintegration and implicit-function-theorem argument give $\|\mu_z-\mu_{z'}\|_{\mathrm{TV}}\le C_1|z-z'|^\beta$, and hence $|\mathbb{E}[p|\Phi_h^{(L)}=z]-\mathbb{E}[p|\Phi_h^{(L)}=z']|\le M_{D,L}C_1|z-z'|^\beta$ by Lemma 5.1. Standard Nadaraya–Watson theory [2] for a Hölder(β) regression function on a $d_{\mathrm{eff}}(L)$-dimensional domain then gives the stated rate $n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))}$ at the optimal bandwidth; the $M_{D,L}^2$ factors cancel in the bias-variance balance and the bound is uniform over $\mathcal{P}_D^{(L)}$. □

Theorem 5.7 (Full convergence $\hat\delta(L,n)\to\delta(L)$). *Under the hypotheses of Proposition 5.6 and with $d>2s$:*

$$\mathbb{E}\big|\hat\delta(L,n)-\delta(L)\big|\ \le\ C\sqrt{D^d/n}\ +\ C_2(D,L)^{1/2}\cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$$

and $P(|\hat\delta(L,n)-\delta(L)|>\varepsilon_n)\to0$ exponentially, where $\varepsilon_n=C'n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L,n)-\delta(L)| \;\le\; \underbrace{|\hat{\delta}(L,n)-\mathbb{E}[\hat{\delta}(L,n)]|}_{\text{concentration, }(\star)} \;+\; \underbrace{|\mathbb{E}[\hat{\delta}(L,n)]-\delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C\sqrt{D^d/n}$ (Lemma 5.4); Source 2 is bounded by $C_2(D,L)^{1/2}\cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$ (Proposition 5.6). The concentration term is bounded by ($\star$) (Proposition 5.2). Combining at $t=\varepsilon_n/2$ gives the probability statement. $\square$

6 The Algebra Theorem

This section assembles the results of Sections 4 and 5 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T.

6.1 Effective dimension and the piecewise rate

Definition 6.1 (Effective dimension)**.** The *effective dimension at lag L* is

$$d_{\mathrm{eff}}(L) \;:=\; \operatorname{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x\in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ-a.e.)

Lemma 6.2 (d_{eff} step function)**.** *Under (G1)–(G2) with X a compact smooth d-manifold and μ smooth positive density:*

$$d_{\mathrm{eff}}(L) \;=\; \min(L+1,\, d) \quad \textit{for all } L\ge 0.$$

Proof. The upper bound $d_{\mathrm{eff}}(L)\le \min(L+1,d)$ is immediate from dimension counting. For the lower bound, the Sard–Smale theorem [9] is applied to two failure maps encoding rank deficiency. In each case, (G1) (distinct orbit points) gives surjectivity of the derivative of the failure map onto the target, making the failure locus a submanifold of too low a dimension to carry μ-mass; (G2) places h in the resulting residual generic set. Full details of the transversality argument follow the template of Takens [11]. $\square$

Proposition 6.3 (Piecewise pre-Takens rate)**.** *Assume (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f\in \mathrm{Hölder}(s)\cap L^\infty(\mu)$, $d>2s$, and $T\in C^r$ with $r\ge 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D=D^*(n,L)$ chosen to balance approximation and statistical terms. Then for each $L\ge 0$:*

$$\big\|f-\hat{f}_n^{(L)}\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty}\cdot\delta(L)^{1/2} \;+\; C(f,d,s)\cdot n^{-s/(2s+d_{\mathrm{eff}}(L))}$$

with probability tending to 1, where $d_{\mathrm{eff}}(L)=\min(L+1,d)$. In the Takens regime ($L\ge d-1$): $d_{\mathrm{eff}}(L)=d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 4.4). The bias term is bounded by $2\|f\|_{L^\infty}\cdot\delta(L)^{1/2}$ (Lemma 4.3). By Lemma 6.2, $\Phi_h^{(L)}(X)$ is a $d_{\mathrm{eff}}(L)$-dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f\mid \mathcal{O}_h^{(L)}](x)=g^*(\Phi_h^{(L)}(x))$ for some $g^*\in\mathrm{Hölder}(s)$; standard polynomial approximation on a $d_{\mathrm{eff}}(L)$-dimensional smooth domain gives approximation error $\le C_{\mathrm{approx}}\cdot D^{-s/d_{\mathrm{eff}}(L)}$. The statistical term is $\le C_{\mathrm{stat}}(D^{d_{\mathrm{eff}}(L)}/n)^{1/2}$ by Proposition 5.2 with d replaced by $d_{\mathrm{eff}}(L)$. Balancing: $D^*\sim n^{d_{\mathrm{eff}}(L)/(2s+d_{\mathrm{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\mathrm{eff}}(L))})$. $\square$

6.2 The oracle lag and the elbow stopping rule

Definition 6.4 (Oracle lag)**.** The *oracle lag* is

$$L^*(n) \;:=\; \min\bigl\{L \geq 0 : \delta(L) \leq C_0 \cdot n^{-2s/(2s+d)}\bigr\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L,n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 6.5 (Elbow stopping rule)**.** Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* \;:=\; \min\bigl\{L \geq 0 : \hat{\delta}(L+1,n) - \hat{\delta}(L,n) > -\tau_n\bigr\}.$$

Theorem 6.6 (Elbow location)**.** *Assume the hypotheses of Proposition 6.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1-e^{-\lambda})Ae^{\lambda}/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then*

$$P\bigl(\hat{L}^* = L^*(n)\bigr) \;\to\; 1 \quad \textit{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n)+1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 5.7, $|\hat{\delta}(L,n) - \delta(L)| \leq \varepsilon_n = C'n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^*+1,n) - \hat{\delta}(L^*,n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L+1) \geq \Delta(L) \geq C_\lambda \cdot C'n^{-2s/(2s+d)}$. Tracking via Theorem 5.7:

$$\hat{\delta}(L,n) - \hat{\delta}(L+1,n) \;\geq\; C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n \;=\; (C_\lambda - 2)C'n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n/C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C'n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 6.7 (Scope of the elbow theorem)**.** The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n / L^*(n) \to 0$ relative to ε_n, so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here.

□

Theorem 6.8 (Algebra Theorem)**.** *Assume:*

(i) Hypotheses of Proposition 6.3: (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f \in$ Hölder$(s) \cap L^\infty(\mu)$, $d > 2s$, $T \in C^r$, $r \geq 2$.

(ii) Exponential mixing: $\delta(L) \leq Ae^{-\lambda L}$ for some $A, \lambda > 0$.

(iii) Resolvability: $C_\lambda > 4$.

(iv) Exponential mixing for the estimator bias (Proposition 5.6) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

Let $\hat{L}^*$ be the elbow stopping rule (Definition 6.5) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then

$$\|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} \;=\; O_p\big(n^{-s/(2s+d)}\big).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for Hölder(s) *functions on a d-dimensional domain [10]. No oracle input (mixing rate, $L^*(n)$, Sobolev index s) is required.*

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 6.6:

$$\begin{aligned}
\|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} &= \|f - \hat{f}_n^{(L^*(n))}\|_{L^2(\mu)} \\
&\le 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&\le 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&= O(n^{-s/(2s+d)}).
\end{aligned}$$

□

6.3 Entropy characterisation of reconstruction

Definition 6.9 (Fibre mixing)**.** Fix $\varepsilon > 0$. An event $S^* \in \mathcal{B}$ is (c,ε)*-mixing across fibres* if there exists $c > 0$ such that

$$\mu_z(S^*)\,\mu_z\big((S^*)^c\big) \;\ge\; c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

Ergodicity of T is expected to force this condition (see Section 9), but the general derivability question remains open; see the open problems below. This condition is independent of (US) and is not a standing assumption here; it is used only in Corollary 6.12 below.

Lemma 6.10 (Positive-fraction balance)**.** *Let $S^* \in \mathcal{B}$ be an ε-near-maximizer of $\delta(L)$, i.e. $\mathbb{E}[\mathrm{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \ge \delta(L) - \varepsilon$. Define the* large-fibre contribution

$$\kappa \;:=\; \int_{\{p_z \ge \varepsilon\}} p_z\, \mu_z(S^*)\, \mu_z\big((S^*)^c\big)\, d\nu_L(z).$$

For any $\eta \in (0, \frac{1}{2})$ with $\kappa > \eta$, the set

$$G(\eta) \;:=\; \big\{z : p_z \ge \varepsilon,\ \mu_z(S^*)\,\mu_z\big((S^*)^c\big) \ge \eta\big\}$$

of ε-large balanced fibres satisfies

$$\nu_L\big(G(\eta)\big) \;\ge\; 4(\kappa - \eta).$$

No dynamical hypothesis is required.

Proof. Partition the ε-large fibres into monochromatic fibres $\mathcal{M} = \{z : p_z \ge \varepsilon,\ \mu_z(S^*)\mu_z((S^*)^c) < \eta\}$ and balanced fibres $G(\eta)$. On $\mathcal{M}$: $\mu_z(S^*)\mu_z((S^*)^c) < \eta$, so

$$\int_{\mathcal{M}} p_z\, \mu_z(S^*)\mu_z\big((S^*)^c\big)\, d\nu_L \;\le\; \eta \int_{\mathcal{M}} p_z\, d\nu_L \;\le\; \eta.$$

Hence $\int_{G(\eta)} p_z \mu_z(S^*)\mu_z((S^*)^c)\, d\nu_L \ge \kappa - \eta$. On $G(\eta)$: $\mu_z(S^*)\mu_z((S^*)^c) \le \frac{1}{4}$ pointwise, so

$$\kappa - \eta \;\le\; \int_{G(\eta)} p_z\, \mu_z(S^*)\mu_z\big((S^*)^c\big)\, d\nu_L \;\le\; \tfrac{1}{4} \int_{G(\eta)} p_z\, d\nu_L \;\le\; \tfrac{1}{4}\, \nu_L\big(G(\eta)\big).$$

□

Lemma 6.11 (Fibre mixing bound)**.** *Let S^* be a $(\delta(L) - \varepsilon)$-near-maximizer of* $\mathrm{E}[\mathrm{Var}(\mathbb{1}_\cdot \mid \mathcal{O}_h^{(L)})]$ *that is (c,ε)-mixing across fibres (Definition 6.9). Then*

$$(\mu \otimes \mu)(R_L) \;\le\; \frac{1}{c}\,\delta(L) + C(c)\,\varepsilon, \qquad C(c) := \frac{1+c}{c}.$$

Proof. Write $(\mu\otimes\mu)(R_L)=\int p_z^2\,d\nu_L(z)$ and split into ε-large and small-fibre parts.

Large fibres ($p_z \geq \varepsilon$). The (c,ε)-mixing condition gives $\mu_z(S^*)\mu_z((S^*)^c) \geq c\,p_z$ on ε-large fibres, so:

$$c\int_{\{p_z\geq\varepsilon\}} p_z^2\,d\nu_L(z) \;\leq\; \int p_z\,\mu_z(S^*)\mu_z\big((S^*)^c\big)\,d\nu_L(z) \;=\; \mathrm{E}\big[\mathrm{Var}(\mathbb{1}_{S^*}\mid \mathcal{O}_h^{(L)})\big] \;\leq\; \delta(L).$$

Hence $\int_{\{p_z\geq\varepsilon\}} p_z^2\,d\nu_L(z) \leq \frac{1}{c}\,\delta(L)$.

Small fibres ($p_z < \varepsilon$). Each such fibre satisfies $p_z^2 \leq \varepsilon\,p_z$, so

$$\int_{\{p_z<\varepsilon\}} p_z^2\,d\nu_L(z) \;\leq\; \varepsilon\int p_z\,d\nu_L(z) = \varepsilon.$$

Combining. Since $C(c)=\frac{1+c}{c}\geq 1$ we have $\varepsilon \leq C(c)\,\varepsilon$, and therefore

$$(\mu\otimes\mu)(R_L) \;\leq\; \frac{1}{c}\,\delta(L) + C(c)\,\varepsilon. \qquad \square$$

The collision probability (2) defines a natural information-theoretic quantity:

$$H_2(\nu_L) \;:=\; -\log\big((\mu\otimes\mu)(R_L)\big) \;=\; -\log\int p_z^2\,d\nu_L(z). \tag{4}$$

This is the *Rényi-2 (collision) entropy* [8] of the delay-vector distribution.

Corollary 6.12 (Entropy characterisation of reconstruction)**.** *Under the fibre mixing condition of Definition 6.9:*

$$\delta(L)\to 0 \qquad\Longleftrightarrow\qquad H_2(\nu_L)\to+\infty.$$

Equivalently, reconstruction occurs if and only if the probability that two independently drawn delay vectors coincide vanishes as $L\to\infty$. Thus reconstruction is equivalent to the vanishing of collision probability $(\mu\otimes\mu)(R_L)$ introduced in Section 3.

Proof. Lemma 4.1 gives $\delta(L) \leq \frac{1}{2}(\mu\otimes\mu)(R_L)$. For the reverse, Lemma 6.11 with a $\delta(L)$-near-maximizer S^* satisfying the (c,ε)-mixing condition gives $(\mu\otimes\mu)(R_L) \leq \frac{1}{c}\,\delta(L)+C(c)\,\varepsilon$, so $\delta(L)\to 0 \iff (\mu\otimes\mu)(R_L)\to 0$. Since $(\mu\otimes\mu)(R_L)=e^{-H_2(\nu_L)}$, this is equivalent to $H_2(\nu_L)\to+\infty$. $\square$

Thus Lemma 6.10 guarantees a positive ν_L-fraction of ε-large fibres are η-balanced without any dynamical hypothesis; the fibre mixing condition is precisely the upgrade from positive-fraction to ν_L-a.e. balance.

Remark 6.13 (Empirical entropy witness)**.** The empirical analogue of $H_2(\nu_L)$ is

$$\hat{H}_2\Big(\nu_L^{(n)}\Big) \;:=\; -\log\left(\frac{1}{n^2}\sum_{i,j=1}^{n}\mathbb{1}\big[\Phi_h^{(L)}(x_i)=\Phi_h^{(L)}(x_j)\big]\right),$$

the negative log of the empirical collision probability. This is the third computable witness for reconstruction. It is *computationally simpler* than $\hat{\delta}(L,n)$: it requires no optimisation over sets, only histogram counts over delay vectors. Under fibre mixing, Corollary 6.12 shows that $\hat{L}^*$ could equivalently be defined as the first L at which $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n.

7 The Dynamics Theorem

7.1 Lipschitz continuity of the delay map

Lemma 7.1 (Upper Lipschitz bound)**.** *Assume $T \in \mathrm{Diff}^r(X)$, $h \in C^r(X)$, $r \geq 1$. Then $\Phi_h^{(L)}$ is Lipschitz:*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\leq\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X,$$

where $C_L := \|Dh\|_{L^\infty} \cdot \big(\sum_{j=0}^{L} \|DT^j\|_{L^\infty}^2\big)^{1/2}$.

Proof. Component j of $\Phi_h^{(L)}$ is $h \circ T^j$. By the chain rule and compactness of X: $|h(T^j x) - h(T^j x')| \leq \|Dh\|_{L^\infty} \cdot \|DT^j\|_{L^\infty} \cdot d_X(x, x')$. Taking Euclidean norm over $j = 0, \ldots, L$ gives the result. □

Lemma 7.2 (Local lower Lipschitz bound)**.** *Under (G1)–(G2) with $L \geq d-1$ (Takens regime), $T \in C^r$, $r \geq 2$: for μ-a.e. $x \in X$, there exists a neighbourhood U_x and a constant $c_x = \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ such that*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\geq\; c_x \cdot d_X(x, x') \quad \forall\, x' \in U_x.$$

Proof. By Lemma 6.2, $D\Phi_h^{(L)}|_x$ has full rank d for μ-a.e. x when $L \geq d-1$, so $\sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ μ-a.e. The lower bound on U_x follows from the inverse function theorem applied to the immersion $\Phi_h^{(L)}$ at x. □

Remark 7.3 (Gap between local and global lower bound)**.** Lemma 7.2 gives a *local* lower bound $c_x > 0$ varying over X. (G1)–(G2) guarantee $\sigma_{\min} > 0$ μ*-a.e.*, but not on all of X. Condition (US) imposes $c := \inf_X \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, which is strictly stronger. The gap is genuine: there exist pairs (T, h) satisfying (G1)–(G2) but violating (US) on a μ-null set.

7.2 Uniform separation

For deterministic T, the conditional regularity kernel specialises to the Dirac family $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$. The TV separation pseudometric is therefore

$$d_L(x, x') \;:=\; \|\Pi_h^{(L)}(x, \cdot) - \Pi_h^{(L)}(x', \cdot)\|_{\mathrm{TV}} \;=\; \mathbb{1}\big[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')\big] \;\in\; \{0, 1\}.$$

Separation is binary: $d_L(x, x') = 1$ if and only if x and x' are distinguished by $\mathcal{O}_h^{(L)}$.

Theorem 7.4 (Uniform separation under (US))**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, $L \geq d-1$. Then $\Phi_h^{(L)}$ is bi-Lipschitz on X:*

$$c \cdot d_X(x, x') \;\leq\; |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\leq\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X.$$

For any estimator $\hat{\Gamma}_h^{(n)}$ with uniform error $|\hat{\Gamma}_h^{(n)}(x) - \Phi_h^{(L)}(x)| \leq \varepsilon_n$ uniformly in x, the empirical separation $\widehat{d}_L(x, x') := |\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')|$ satisfies

$$\widehat{d}_L(x, x') \;\geq\; c \cdot d_X(x, x') - 2\varepsilon_n.$$

In particular, if $d_L(x, x') = 1$ (i.e. $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$), then $\widehat{d}_L(x, x') > 0$ for all n large enough that $\varepsilon_n < c \cdot d_X(x, x')/2$.

Proof. Upper bound: Lemma 7.1. Lower bound: (US) directly gives $|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c \cdot d_X(x, x')$ for all $x, x' \in X$. For the empirical bound: by the triangle inequality, $|\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')| \;\geq\; |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| - 2\varepsilon_n \;\geq\; c \cdot d_X(x, x') - 2\varepsilon_n$. If $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ then by bi-Lipschitz $d_X(x, x') \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')|/C_L > 0$, so the right side is eventually positive. □

Theorem 7.5 (Dynamics Theorem)**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, μ smooth positive density on compact d-manifold X, $L \geq d-1$. Let $\hat{\Gamma}_h^{(n)}$ be the empirical delay map estimator with uniform error $\varepsilon_n = C'' n^{-\beta/(2\beta+d)}$ (Proposition 5.6). Then:*

(a) (Convergence) $\|\hat{\Gamma}_h^{(n)} - \Phi_h^{(L)}\|_{L^\infty} \to 0$ *at rate* $n^{-\beta/(2\beta+d)}$.

(b) (Separation) *For $\mu \otimes \mu$-a.e. (x, x') with $x \neq x'$: $\widehat{d}_L(x, x') > 0$ for all large n.*

(c) (Scope) *Condition (US) is not a consequence of (G1)–(G2). Without (US), the kernel witness $\widehat{d}_L$ can vanish on a positive-measure set of pairs even when $d_L = 1$ and reconstruction holds.*

Proof. (a) Proposition 5.6 with $d_{\mathrm{eff}}(L) = d$ gives $\varepsilon_n \to 0$ at rate $n^{-\beta/(2\beta+d)}$.

(b) Since μ is non-atomic (smooth positive density), $\mu \otimes \mu$ almost every pair satisfies $d_X(x, x') > 0$. For such pairs $d_L(x, x') = 1$ (reconstruction and injectivity of $\Phi_h^{(L)}$), and by Theorem 7.4, $\widehat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n$. For fixed $x \neq x'$, $c \cdot d_X(x, x') > 0$ and $\varepsilon_n \to 0$, so the right side is eventually positive.

(c) The proof of Theorem 7.4 uses (US) in the lower bound. Without it, there exist points x where $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ along sequences $x'_n \to x$, and $|\Phi_h^{(L)}(x'_n) - \Phi_h^{(L)}(x)|/d_X(x, x'_n) \to 0$. Any estimator inherits this degeneracy. □

8 The Conjunction Theorem

This section formalises the complete reconstruction diagnostic and proves that both witnesses certify the same event when reconstruction and (US) hold.

8.1 The honest bridge

The delay vector $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$ is a trajectory of the Markov chain on h-space with one-step kernel Π_h. The σ-algebra $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ is exactly the information generated by L steps of this Markov chain starting from x. This gives an algebraic identity:

$$\sigma\big(\Phi_h^{(L)}\big) \;=\; \mathcal{O}_h^{(L)},$$

which is not an approximation but a definition. Consequently, the σ-algebra separation event and the metric separation event are identical:

Lemma 8.1 (Algebra separation equals metric separation)**.** *For deterministic T and $\mu \otimes \mu$-a.e. (x, x'):*

$$d_L(x, x') = 1 \iff x \text{ and } x' \text{ are separated by } \mathcal{O}_h^{(L)}.$$

These are the same event, not merely correlated.

Proof. For deterministic T: $\Pi_h^{(L)}(x, \cdot\,) = \delta_{\Phi_h^{(L)}(x)}$, so $d_L(x, x') = \|\delta_{\Phi_h^{(L)}(x)} - \delta_{\Phi_h^{(L)}(x')}\|_{\mathrm{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')]$. And x is separated from x' by $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ if and only if $\exists E \in \mathcal{O}_h^{(L)}$ with $\mathbb{1}_E(x) \neq \mathbb{1}_E(x')$, i.e., if and only if $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ (since $\mathcal{O}_h^{(L)}$ is generated by the level sets of $\Phi_h^{(L)}$). The two conditions are identical. □

Under reconstruction ($\delta(L) = 0$, equivalently $\Phi_h^{(L)}$ injective μ-a.e.) and (US), injectivity holds everywhere, so $d_L(x, x') = 1$ for $\mu \otimes \mu$-a.e. $x \neq x'$.

8.2 The joint diagnostic

Definition 8.2 (Joint reconstruction diagnostic)**.** Fix tolerances $\tau_n, \eta_n > 0$. Define

$$\mathrm{Reconstruct}(L,n) \ := \ \big\{\hat{\delta}(L,n) \le \tau_n\big\} \ \wedge \ \big\{\widehat{d}_L(x,x') > \eta_n \text{ for } \mu\otimes\mu\text{-most pairs}\big\}.$$

Theorem 8.3 (Conjunction Theorem)**.** *Assume the hypotheses of Theorem 6.8 (for the algebra side) and Theorem 7.5 (for the dynamics side). Set $\tau_n = \frac{C_\lambda}{2}C'n^{-2s/(2s+d)}$ and $\eta_n = \sqrt{\varepsilon_n}$ where $\varepsilon_n = C'n^{-2s/(2s+d)}$. Then:*

(a) (Conjunction fires under reconstruction + (US)). *Under $\delta(L) = 0$ for $L \ge L^*(n)$ and (US):*

$$P\big(\mathrm{Reconstruct}(\hat{L}^*, n)\big) \ \to \ 1 \quad \textit{exponentially in } n.$$

(b) (Algebra witness fails when reconstruction fails). *If $\delta(L) > 0$ for all L: $P(\hat{\delta}(L,n) \le \tau_n) \to 0$ for every fixed L, and $\hat{L}^* \to \infty$.*

(c) (Kernel witness fails when (US) fails). *If $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ on a positive-measure set S: $P(\widehat{d}_L > \eta_n$ for $\mu\otimes\mu$-most pairs$) \to 0$ even when $\hat{\delta}(L,n) \to 0$.*

In particular: the conjunction is not achievable when either hypothesis fails.

Proof. Part (a). By Theorem 6.6, $P(\hat{L}^* = L^*(n)) \to 1$ exponentially. On $\{\hat{L}^* = L^*(n)\}$: (i) $\hat{\delta}(\hat{L}^*, n) \le \tau_n$ whp by Theorem 5.7 and the definition of $L^*(n)$ ($\delta(L^*(n)) = 0$ under reconstruction). (ii) By Theorem 7.5(b), $\widehat{d}_{\hat{L}^*}(x,x') > 0$ for $\mu\otimes\mu$-a.e. (x,x') with $x \ne x'$, eventually. The set $\{d_X(x,x') \le (2\varepsilon_n + \eta_n)/c\}$ has $\mu\otimes\mu$-measure $\to 0$ as $\eta_n/c \to 0$, so $\widehat{d}_{\hat{L}^*} > \eta_n$ for $\mu\otimes\mu$-most pairs.

Part (b). If $\delta(L) > 0$ for all L, then by Theorem 5.7, $\hat{\delta}(L,n) \ge \delta(L) - \varepsilon_n \ge \delta(L)/2 > \tau_n$ with high probability for all fixed L, so the first condition of Reconstruct never fires.

Part (c). If $S = \{x : \sigma_{\min}(D\Phi_h^{(L)}|_x) = 0\}$ has $\mu(S) > 0$, then for $x \in S$ there exist sequences $x'_k \to x$ with $|\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)|/d_X(x'_k, x) \to 0$. Any consistent estimator satisfies $\widehat{d}_L(x,x'_k) \le |\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)| + 2\varepsilon_n \to 0$. The set of pairs where $\widehat{d}_L \le \eta_n$ thus has positive $\mu\otimes\mu$-measure, and the kernel condition fails. □

9 Discussion

The results of this paper divide into two layers. Section 2 establishes general observational certification: the population separation defect $\delta(\mathcal{G})$ converges to zero under CE and honest refinement, the empirical U-statistic proxy concentrates under IID sampling, and a persistent empirical floor is evidence against CE. These results require no temporal or geometric structure. Sections 6–8 establish the dynamical sharpening: honest refinement is automatic under lag growth, and mixing upgrades consistency to minimax-optimal rates with computable witnesses. The open problems below all concern the dynamical layer; the general layer is complete at the level of consistency.

Polynomial mixing. The Algebra Theorem (Theorem 6.8) and the Elbow Location Theorem (Theorem 6.6) require exponential mixing with resolvability constant $C_\lambda > 4$. Under polynomial mixing $\delta(L) \lesssim L^{-\alpha}$, the elbow drop at $L^*(n)$ satisfies $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \to 0$ relative to ε_n: the elbow and the noise floor merge asymptotically and cannot be separated. A different stopping criterion (e.g. integrated elbow, or a bandwidth-adaptive rule) is needed.

The $d > 2s$ hypothesis. Proposition 5.2 requires $d > 2s$ for the Rademacher concentration to be useful at rate ε_n. When $d \le 2s$, localised Rademacher complexity (Bartlett–Bousquet–Mendelson [1]) gives adaptive concentration without this condition but requires additional work.

Uniform separation from first principles. Condition (US) is a hypothesis in Theorems 7.5 and 8.3. The natural conjecture is that T Anosov with h generic implies (US) via the uniform hyperbolicity constants. This would close the gap between (G1)–(G2) and (US) in the hyperbolic category.

Remark 9.1 (Observable separation and Lyapunov exponents)**.** The bi-Lipschitz property of $\Phi_h^{(L)}$ established in Theorem 7.5 under (G1)–(G2)+(US) has a direct bearing on the observable separation exponent introduced in [5] (Remark II.7). Recall that the *observable separation exponent* is

$$\lambda_h(x, y) \; := \; \limsup_{n\to\infty} \frac{1}{n} \log d\big(\Phi_h^{(L)}(T^n x), \, \Phi_h^{(L)}(T^n y)\big),$$

measuring how fast the observable distinguishes initial conditions under iteration. The bi-Lipschitz sandwich $c\, d(x, y) \leq d(\Phi_h^{(L)}(x), \Phi_h^{(L)}(y)) \leq C\, d(x, y)$, valid μ-a.e. under (G1)–(G2)+(US), implies

$$\lambda_h(x, y) \; = \; \lambda_1(x, y) \quad \mu\text{-a.e.},$$

where λ_1 is the top Lyapunov exponent of T in the ambient metric. That is, under faithful reconstruction and uniform separation, the observable separation rate and the geometric instability rate coincide. This identifies the observable separation exponent as a computable proxy for λ_1 accessible directly from time-series data, without knowledge of the tangent bundle or the derivative cocycle.

Stochastic T. For stochastic dynamics, $\Pi_h^{(L)}(x, \cdot\,)$ is not a Dirac delta. The TV separation d_L is no longer binary, and Hölder continuity of $x \mapsto \Pi_h^{(L)}(x, \cdot\,)$ in TV is both necessary and non-trivial. The four-link chain of Proposition 5.6 adapts, but the Conjunction Theorem's proof structure changes substantially.

Sharp rates for the data-driven lag. Theorem 6.6 shows $P(\hat{L}^* = L^*(n)) \to 1$ but does not bound the fluctuations of $\hat{L}^*$ around $L^*(n)$ on the event $\{\hat{L}^* \neq L^*(n)\}$. A Berry–Esseen-type result for $\hat{L}^* - L^*(n)$ would sharpen the stopping rule.

Concentration of the entropy witness. The empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ is a computationally simpler stopping criterion (Remark 6.13). Asymptotic equivalence with $\hat{\delta}(L, n)$ is established under fibre mixing (Corollary 6.12), but a finite-n concentration result for $\hat{H}_2(\nu_L^{(n)})$ — comparable to Proposition 5.2 for $\hat{\delta}(L, n)$ — is not proved here. Such a result requires only McDiarmid-type bounds on the U-statistic $\frac{1}{n^2} \sum_{i,j} \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]$, with no new structural assumptions beyond those already made.

Derivability of fibre mixing (the main conceptual open problem). The central open question is whether the fibre mixing condition (Definition 6.9) is derivable from dynamical hypotheses, or whether it is an irreducible admissibility condition analogous to CE in [6].

Lemma 6.10 settles Step A: a positive ν_L-fraction of ε-large fibres are η-balanced, with an explicit lower bound depending only on the large-fibre contribution κ and not on any dynamical hypothesis.

The open question is Step B: what is the weakest dynamical condition that upgrades positive-fraction balance to ν_L-a.e. balance? Ergodicity is the natural candidate: in an ergodic system, a near-maximizer of $\delta(L)$ must cross fibres rather than align with them, since aligned events are $\mathcal{O}_h^{(L)}$-measurable and thus carry zero approximation error. The informal argument is clear, but formalizing the connection between ergodicity and the a.e. balance condition is the isolated open gap. If ergodicity suffices, the fibre mixing condition is derivable and the structural analogy with CE breaks; if it does not, the analogy is complete.

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